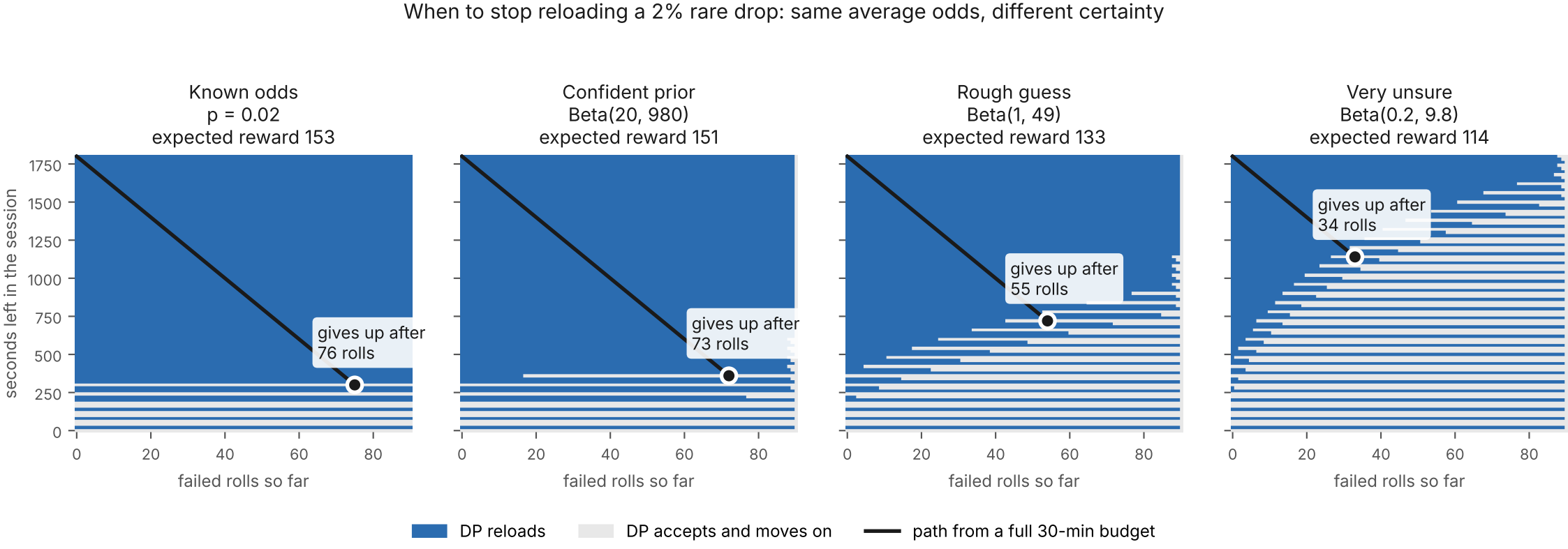
When to stop reloading a 2% rare drop: same average odds, different certainty
Known odds
p = 0.02
expected reward 153
Confident prior
Beta(20, 980)
expected reward 151
Rough guess
Beta(1, 49)
expected reward 133
Very unsure
Beta(0.2, 9.8)
expected reward 114
seconds left in the session
1750
1500
1250
1000
750
500
250
0
gives up after
76 rolls
gives up after
73 rolls
gives up after
55 rolls
gives up after
34 rolls
0
20
40
60
80
failed rolls so far
DP reloads
DP accepts and moves on
path from a full 30-min budget